

"This is the first time a flexible supercapacitor including all its components has been produced by 3D printing," said Milad Areir at Brunel's Cleaner Electronics Research Group. "The most popular way to produce them is screen printing, but with that you can't print the frame of the supercapacitor on silicone."



3D-printed wearable battery (Source: Brunel University London)

The work, published in *Materials Science and Engineering*, shows the power wristband can be made using cheap products from a household shop instead of sophisticated expensive metals or semiconductors. What's more, these wristbands stand up to stress tests without losing power.

Inductive charging coming to cars soon

Hassle of cords is one of the barriers slowing electric vehicle (EV) adoption. Wireless charging helps EVs surpass the convenience of gas cars. Fortunately, wireless inductive charging for cars may become reality sooner than most expect. At this year's *Paris Motor Show*, both VW



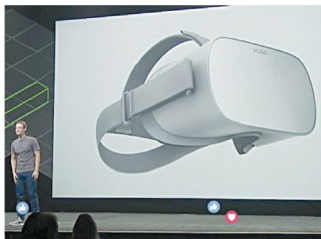
Wireless charging coming soon to an electric vehicle near you (Image courtesy: <https://www.greentechmedia.com/articles/read/wireless-charging-coming-to-an-electric-vehicle-near-you>)

and Mercedes-Benz featured inductive charging in their EV announcements.

This is how vehicle inductive charging works: A length of copper wire acting as the first conductor is coiled around a piece of ferrite (a substance made of oxides of iron and other metals), which amplifies the magnetic field generated. This arrangement is housed in a flat case to create a pad that is easily buried. When a vehicle equipped with a suitable 'pickup' coil stops or parks above this device, and alternating current is fed into the pad, a similar current is induced in the pickup. This is then converted into direct current by a rectifier, and is used to top up the vehicle's battery.

\$199 VR headset that needs no phone

Facebook CEO Mark Zuckerberg has unveiled a new mobile headset called 'Oculus Go' that lets users effortlessly enter virtual reality (VR) with no PC or wires attached. Unlike the Oculus Rift, which needs to be tethered to a PC, or Sam-



Facebook CEO Mark Zuckerberg unveiling the Oculus Go mobile headset (Photo by James Martin/CNET)

sung's Gear VR, which requires a phone, the Oculus Go is an all-in-one VR headset made to fit you. It has cameras on the headset and uses computer vision technology for orientation-tracking. The device is said to be designed with breathable fabrics, adjustable straps and the company's best lenses yet. And it can even be used with glasses.

Crystal-clear optics, integrated spatial audio and optimised 3D graphics come together to bring your virtual world to life with control in the palm of your hand. So draw constellations in the night sky, cast a fishing line with precision, capture enemies in a multiplayer standoff, and more.

The headset starts at \$199 and is going to ship early next year. It utilises a fast-switch WQHD LCD screen and will be cross-compatible with all Gear VR titles.